

# Docker Networking

## To begin with the Lab

### Summary of the Lab

This lab covered Docker networking, file management, and bind mounts. A custom Docker network was created and a container was connected to it. Files were copied between the host and container using `docker cp` and verified. Finally, a bind mount was configured to share files between the host and container, allowing real-time updates to appear in the web application without manually copying files.

- Let's list the docker network

```
sudo docker network ls
```

```
ubuntu@ip-172-31-45-84:~$ sudo docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
60b010e5e5c3        bridge             bridge              local
c65eca012e95        host               host                local
6ef111b3abf5        none               null                local
ubuntu@ip-172-31-45-84:~$
```

- Let's create a new network

```
sudo docker network create app-network
```

- then list the network

```
sudo docker network ls
```

```
ubuntu@ip-172-31-45-84:~$ sudo docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
60b010e5e5c3        bridge             bridge              local
c65eca012e95        host               host                local
6ef111b3abf5        none               null                local
ubuntu@ip-172-31-45-84:~$ sudo docker network create app-network
525c457a27ab280cfde89872110b4c255ed09cd39754c503042ce9dbc4525e1d
ubuntu@ip-172-31-45-84:~$ sudo docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
525c457a27ab        app-network         bridge              local
60b010e5e5c3        bridge             bridge              local
c65eca012e95        host               host                local
6ef111b3abf5        none               null                local
ubuntu@ip-172-31-45-84:~$
```

- Let's attach it to the 'quizzical\_gould' container

```
ubuntu@ip-172-31-45-84:~$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
45d87d140052   php-app:v1 "docker-php-entrypoi..." About a minute Up About a minute 0.0.0.0:80->80/tcp, [::]:80->80/tcp quizzical_gould
ubuntu@ip-172-31-45-84:~$
```

```
sudo docker network connect app-network quizzical_gould
```

- Verify the container is connected

```
sudo docker network inspect app-network
```

```

},
"ConfigOnly": false,
"Options": {},
"Labels": {},
"Containers": {
  "45d87d1400521772bf4b54beda2b00a2ba67572e7a86dff6214a472b119f2dd5": {
    "Name": "quizzical_gould",
    "EndpointID": "925ad712c0ea6ee416f9b91760c3a234fac6f54d3a3125ee7e77f1ca57e8c421",
    "MacAddress": "ca:57:d8:d6:0e:a1",
    "IPv4Address": "172.18.0.2/16",
    "IPv6Address": ""
  }
}
}

```

- Create a file on the EC2 host

```
echo "Welcome to Docker Lab" > sample.txt
```

```
cat sample.txt
```

- Expected output:
- Welcome to Docker Lab

```

ubuntu@ip-172-31-45-84:~$ echo "Welcome to Docker Lab" > sample.txt
cat sample.txt
Welcome to Docker Lab
ubuntu@ip-172-31-45-84:~$ █

```

- Copy the file into the container

```
sudo docker cp sample.txt employee-portal-container:/var/www/html/
```

- If there is no error, the copy was successful.

```

ubuntu@ip-172-31-45-84:~/PHP$ sudo docker ps
CONTAINER ID   IMAGE               COMMAND                  CREATED        STATUS        PORTS                               NAMES
b2dc80a1f529   employee-portal:v1  "docker-php-entrypoi..." About a minute ago Up 59 seconds  0.0.0.0:80->80/tcp, [::]:80->80/tcp  employee-portal-container
ubuntu@ip-172-31-45-84:~/PHP$ sudo docker cp sample.txt employee-portal-container:/var/www/html/
Successfully copied 2.05kB to employee-portal-container:/var/www/html/
ubuntu@ip-172-31-45-84:~/PHP$ █

```

- Verify it inside the container

```
sudo docker cp sample.txt employee-portal-container:/var/www/html/
```

- Then run:

```
ls /var/www/html
```

- You should now see sample.txt.
- Next:

```
cat /var/www/html/sample.txt
```

- Expected output:
- Welcome to Docker Lab

```
ubuntu@ip-172-31-45-84:~/PHP$  
ubuntu@ip-172-31-45-84:~/PHP$  
ubuntu@ip-172-31-45-84:~/PHP$  
ubuntu@ip-172-31-45-84:~/PHP$ sudo docker exec -it employee-portal-container bash  
root@b2dc80alf529:/var/www/html# ls  
Dockerfile index.php sample.txt  
root@b2dc80alf529:/var/www/html# cat sample.txt  
Welcome to Docker Lab  
root@b2dc80alf529:/var/www/html#
```

- Exit the container:

```
exit
```

- Go to the browser and check

### Hello from Cloudfrecks!

Current Time: 2026-07-09 11:27:04

PHP Version: 8.2.32

#### Contents of sample.txt

Welcome to Docker Lab

- Modify the file on the host

```
echo "Docker Copy Command Demo" > sample.txt
```

- Copy the updated file into the container

```
sudo docker cp sample.txt employee-portal-container:/var/www/html/
```

```
ubuntu@ip-172-31-45-84:~$ cd PHP/  
ubuntu@ip-172-31-45-84:~/PHP$ echo "Docker Copy Command Demo" > sample.txt  
ubuntu@ip-172-31-45-84:~/PHP$ sudo docker cp sample.txt employee-portal-container:/var/www/html/  
Successfully copied 2.05kB to employee-portal-container:/var/www/html/  
ubuntu@ip-172-31-45-84:~/PHP$
```

- Verify the updated content
- Refresh the browser.
- Confirm that **Contents of sample.txt** now displays:
- Docker Copy Command Demo

### Hello from Cloudfrecks!

Current Time: 2026-07-09 11:28:39

PHP Version: 8.2.32

#### Contents of sample.txt

Docker Copy Command Demo

- Docker Bind Mounts
- Stop and remove the existing container

```
sudo docker stop employee-portal-container
```

```
sudo docker rm employee-portal-container
```

- Verify:

```
sudo docker ps -a
```

```
ubuntu@ip-172-31-45-84:~/PHP$ sudo docker stop employee-portal-container
employee-portal-container
ubuntu@ip-172-31-45-84:~/PHP$ sudo docker rm employee-portal-container
employee-portal-container
ubuntu@ip-172-31-45-84:~/PHP$ sudo docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
ubuntu@ip-172-31-45-84:~/PHP$ cd ..
ubuntu@ip-172-31-45-84:~$
```

- Create a directory
- Move to your home directory:

```
cd ~
```

- Create a folder:

```
mkdir docker-bind
```

- Move into it:

```
cd docker-bind
```

```
ubuntu@ip-172-31-45-84:~$ mkdir docker-bind
ubuntu@ip-172-31-45-84:~$ cd docker-bind
ubuntu@ip-172-31-45-84:~/docker-bind$
```

- Create index.php

```
nano index.php
```

- Paste:

```
<?php
echo "<h1>Hello from Cloudfrecks!</h1>";
echo "<h2>Docker Bind Mount Demo</h2>";
echo "<p>Current Time: " . date("Y-m-d H:i:s") . "</p>";

echo "<h3>Contents of sample.txt</h3>";

if (file_exists("sample.txt")) {
    echo "<pre>";
    echo htmlspecialchars(file_get_contents("sample.txt"));
    echo "</pre>";
} else {
    echo "<p>sample.txt not found.</p>";
}
```

```
}  
?>
```

- Create sample.txt

```
echo "Welcome to Docker Bind Mount Demo" > sample.txt
```

- Verify:

```
cat sample.txt
```

- Output:
- Welcome to Docker Bind Mount Demo

```
ubuntu@ip-172-31-45-84:~/docker-bind$  
ubuntu@ip-172-31-45-84:~/docker-bind$ echo "Welcome to Docker Bind Mount Demo" > sample.txt  
ubuntu@ip-172-31-45-84:~/docker-bind$ cat sample.txt  
Welcome to Docker Bind Mount Demo  
ubuntu@ip-172-31-45-84:~/docker-bind$
```

- Run the container using a Bind Mount
- Since your image is employee-portal:v1, run:

```
sudo docker run -d --name employee-portal-container --network app-network -p 80:80 --mount type=bind,source=$(pwd),target=/var/www/html employee-portal:v1
```

- Here:
- source=\$(pwd) → Current directory on the EC2 host (~/docker-bind)
- target=/var/www/html → Web root inside the container

```
ubuntu@ip-172-31-45-84:~/docker-bind$ sudo docker run -d --name employee-portal-container --network app-network -p 80:80 --mount type=bind,source=$(pwd),target=/var/www/html employee-portal:v1  
83bf772f23187ccc0c0bec90b697c5173bbe68313e68adee852d57cb567694  
ubuntu@ip-172-31-45-84:~/docker-bind$ sudo docker ps  
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES  
83bf772f2318   employee-portal:v1   "docker-php-entrypoi..."  21 seconds ago   Up 20 seconds   0.0.0.0:80->80/tcp, [::]:80->80/tcp   employee-portal-container  
ubuntu@ip-172-31-45-84:~/docker-bind$
```

- Open the application
- Visit:
- <http://<EC2-Public-IP>/>
- You should see:

---

## Hello from Cloudfreeks!

### Docker Bind Mount Demo

Current Time: 2026-07-09 11:43:22

#### Contents of sample.txt

Welcome to Docker Bind Mount Demo

- Modify sample.txt
- On the EC2 host:

```
echo "Docker Bind Mount Updated Successfully" > sample.txt
```

```
ubuntu@ip-172-31-45-84:~/docker-bind$  
ubuntu@ip-172-31-45-84:~/docker-bind$  
ubuntu@ip-172-31-45-84:~/docker-bind$  
ubuntu@ip-172-31-45-84:~/docker-bind$ sudo docker ps  
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS                               NAMES  
83bf772f2318   employee-portal:vl                 "docker-php-entrypoi..." 3 minutes ago  Up 3 minutes  0.0.0.0:80->80/tcp, [::]:80->80/tcp  employee-portal-container  
ubuntu@ip-172-31-45-84:~/docker-bind$ echo "Docker Bind Mount Updated Successfully" > sample.txt  
ubuntu@ip-172-31-45-84:~/docker-bind$
```

- Refresh the browser.
- You should immediately see:
- Docker Bind Mount Updated Successfully
- No docker cp command required.

## Hello from Cloudfreeks!

### Docker Bind Mount Demo

Current Time: 2026-07-09 11:46:37

#### Contents of sample.txt

Docker Bind Mount Updated Successfully